



An AI protein check for the age we all get to (71+)

Why it starts long before then.

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Agenda

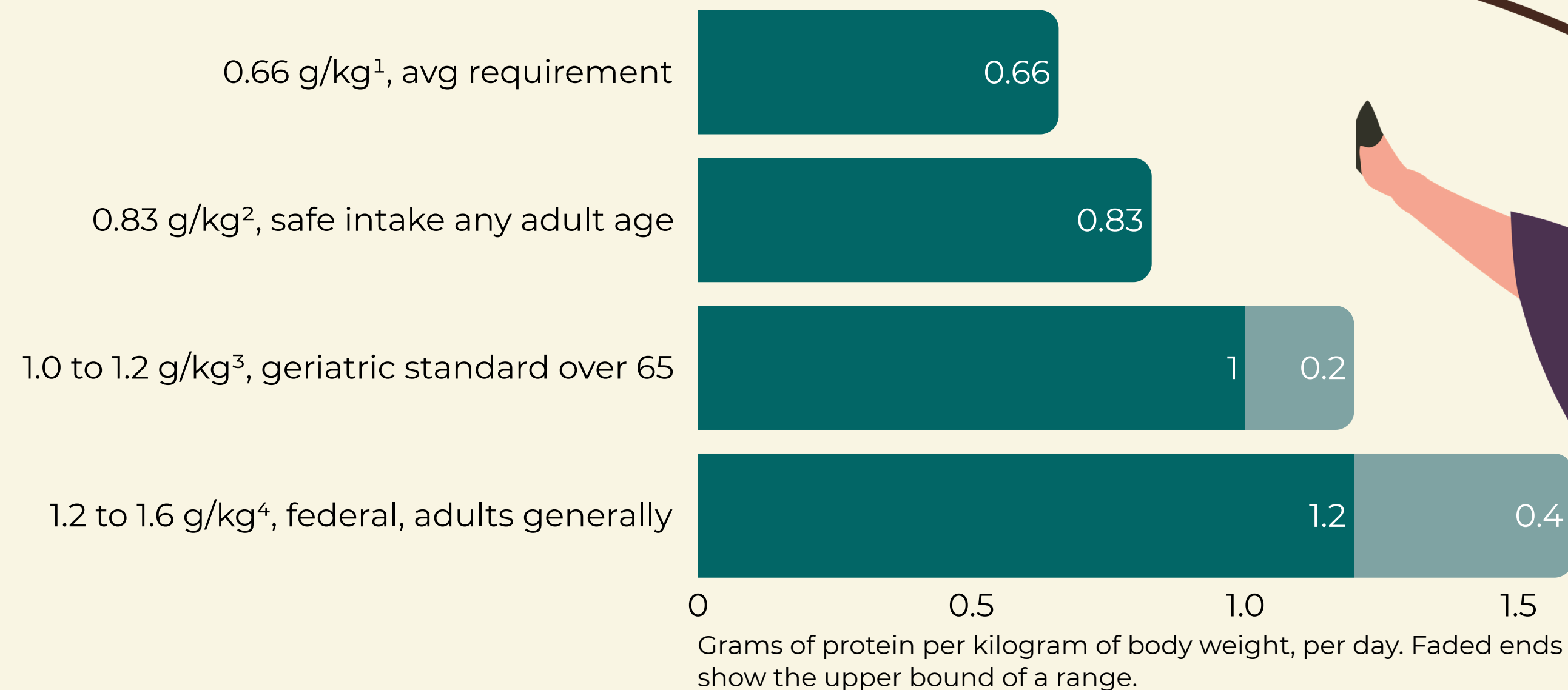
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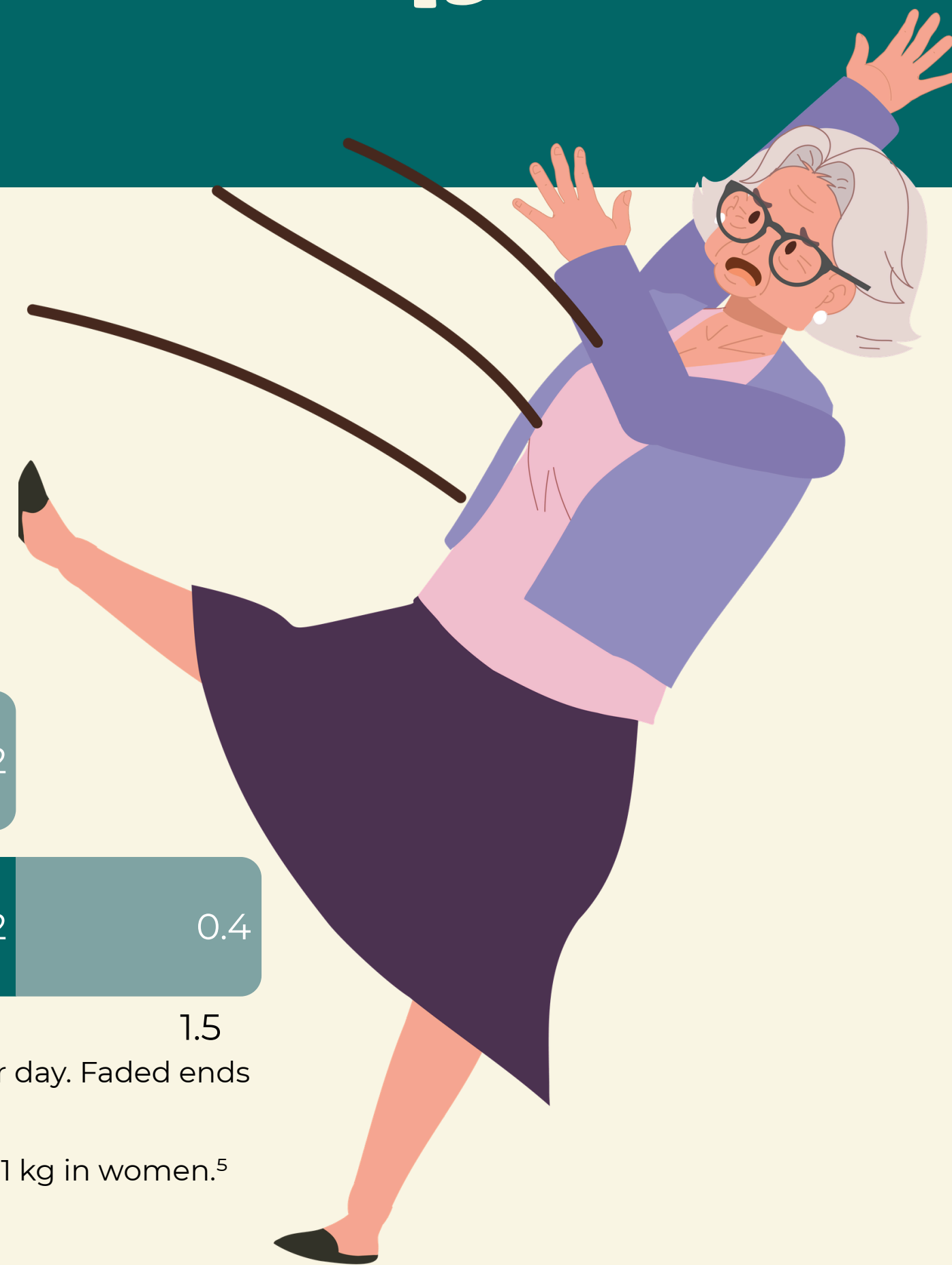
03 Muscle declines every decade after 45

Falls, fractures, and lost independence follow.

Four published protein targets. This tracker uses 1.0 to 1.2 g/kg.

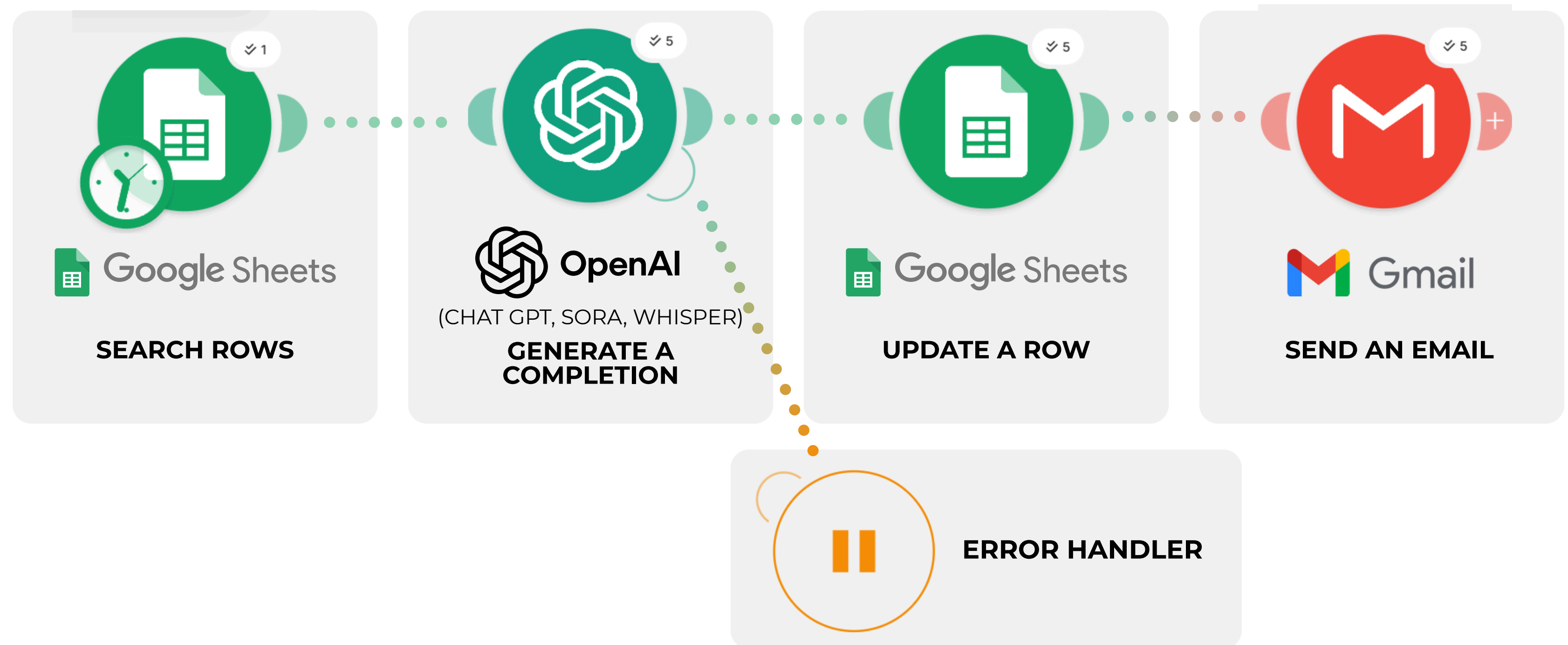


Muscle decline: about 1.9 kg per decade in men, 1.1 kg in women.⁵



04 AI protein tracker

An automated workflow in Make that turns what a senior ate all day into a summary of how much of it was protein.



05 Four tools and Claude for research

The tools run once I start them.

Five steps, one OpenAI call

-  The AI does this.
-  Make, Sheets, and Gmail do this.



Target is a gray step: spreadsheet formula, not the AI.

Your weight in pounds is divided by 2.20462, the number of pounds in a kilogram. Bobby, 181 pounds, is 82 kilograms. At one gram per kilogram his floor is that same number, 82 grams. His goal is 82 times 1.2, rounded, which is 98 grams.

* Claude Not part of the running workflow.

06 RISE

One prompt. Multiple versions. Reliable by the last one.

R

Role:

A kind nutritionist. Never shames.
Kidney questions go to the doctor.

I

Input:

Name, protein targets, kidney flag, the
day's meals.

S

Steps:

Screen. Tally one food per line. Verdict.
One of two closings.

E

Expectation:

Three sentences or fewer. Second
person. A banned-words list.

What broke, v1 to v2

The model added the grams
in its head and miscounted.

What fixed it, v3

Every food on its own line, with a
running total.

More fixes, v4 to v14

Branch logic, prompt leak,
empty-day note, and kidney
safety wording.

07 Project deliverable & live demo

Pick any items, separate with commas, and leave a meal blank if you skipped it.



**Live demo
shown in class**

**A day of meals in, a
warm summary out.
No judgment.**

Eggs and dairy

Large egg (or two eggs, or a three-egg omelet)
Greek yogurt, plain, 1 cup
Cottage cheese, lowfat, 1 cup
Cheese, 1 oz
Cream cheese, 1 oz

Meat, fish and tofu

Chicken breast, cooked, 3 oz
Beef steak, cooked, 3 oz
Pork chop, 4 oz
Turkey breast, 3 oz
Salmon, cooked, 3 oz
Tofu, three-quarters cup

Sandwiches

Turkey sandwich
Turkey and cheese sandwich
Tuna sandwich
Peanut butter and jelly sandwich
Hummus with pita

Bread and pasta

Bread, 2 slices
Toast, 1 slice
Dinner roll
Bagel
Garlic bread
Spaghetti, one serving

Hot dishes and sides

Meatballs (4)
Marinara sauce
Soup, 1 cup
Baked potato
Mashed potatoes

Vegetables and salad

Asparagus
Green beans
Steamed broccoli
Roasted carrots
Green salad

Fruit

Apple
Banana
Tangerine
Watermelon
Fruit cup

Snacks

Walnuts, 1 oz
Energy bar
Donut
Chips, 1 bag
Candy bar
Cookie

Drinks

Milk, 1 cup
Orange juice, 1 cup
Coffee (or coffee with milk)
Tea
Water

Sodas

Soda
Coke
Diet Coke

08 Trials & tribulations

The research and the prompt were the hardest parts.

The problem

- **The data was hard to analyze**

Government data is not collected every year, and the WHO figure is from 2007.

- **The prompt leaked into the note**

Entire sentences of my prompt came back as prose in the note.

What I did

- Confirmed 2007 was the current figure, then chose geriatric data as most relevant to older adults.
- Rewrote the closing-sentence instruction and banned eight words, including suggestion, floor, target, and goal.

09 The best parts

I liked the research, the encouragement, and the moment it worked.

The research

I am a former reporter. Sorting four published standards into one defensible target was the part I enjoyed most.

Designing for encouragement

Every tally note includes a suggestion, encouragement, or congratulations. Never a scold, never a score.

Seeing it work

Meals in, a caring message out, through a chain I built. It runs unattended, but keeping it good is the work.

**I made this for seniors. Muscle starts dropping at 45,
so some of you are already there.**

10 What I learned

Language to the model. Arithmetic to the spreadsheet. Judgment left to me.

Where did AI help me, and where didn't it?

Right for messy food descriptions and warm writing. Wrong for the math, which I kept in fixed logic in the Google Sheet.

Did AI change how I worked?

It gave me an extra layer of checking. I still checked and double-checked everything myself. Every decision was mine.

How did I handle ethics?

No real health data, only fictional people. A kidney screen stops the run.

What can't the tracker do?

It can't reach a food database. One nutrient, about 50 foods held in the prompt. It's a prototype, not a product.

What would I fix before shipping?

Add an API to a food database, expand beyond protein, and put a clinician in the loop.

Where else could I see this working?

Anywhere people need guidance from structured data.

11 Don't forget to eat well at any age

The details behind every number on the slides before this.

Model
gpt-4o-mini

Prompt
RISE, v14

Targets
lb to kg
Floor × 1.0
Goal × 1.2

Food values
About 50,
from USDA via
MyFoodData⁶

Testing
8 fictional
personas

Safety screen
Kidney flag
halts the run

How the model got there

Donut = 3g (running total: 3)

Coffee with milk = 1g (running total: 4)

nine more lines, one food each

Total 24 vs floor 82

Verdict: BELOW FLOOR. Use Step 5a.

What the model wrote

Bobby, you reached a total of 24 grams today, while a good day for you is at least 82 grams. I encourage you to explore ways to increase your protein intake, as it can be beneficial for your overall health. You might want to add a handful of walnuts or almonds to breakfast.

Sources

1. USDA, Agricultural Research Service, 2023. Usual Nutrient Intake, Table A3, p. 3.
2. World Health Organization. WHO Technical Report Series 935, 2007, p. 242.
3. Deutz NEP, Bauer JM, Barazzoni R, et al. Clinical Nutrition 33 (2014), p. 931.
4. HHS and USDA. Dietary Guidelines for Americans, 2025-2030, page 2.
5. Janssen I, Heymsfield SB, Wang Z, Ross R. J Appl Physiol 89: 81-88, 2000, pp. 83 and 87.
6. USDA, Agricultural Research Service. FoodData Central. Values retrieved via MyFoodData, <https://tools.myfooddata.com/nutrition-facts>. Accessed August 5, 2026.

